Indices, Surds and Polynomials

ANSWER KEY

Section 1: Fractional Indices and Exponential Equations

1 Evaluate the exact numerical value of $16^{-\frac{3}{4}} \times 27^{\frac{2}{3}}$.

a. $\frac{9}{8}$ ✓

b. $\frac{8}{9}$

c. $\frac{3}{4}$

d. $\frac{27}{16}$

$16^{-3/4} = (16^{1/4})^{-3} = 2^{-3} = 1/8$. $27^{2/3} = (27^{1/3})^2 = 3^2 = 9$. Product $= 9/8$.

2 Solve the exponential equation $2^{2x+1} - 9(2^x) + 4 = 0$ for all real x.

a. $x = -1$ **and** $x = 2$ ✓

b. $x = 1$ and $x = 4$

c. $x = 0$ and $x = 2$

d. $x = -2$ and $x = 1$

Let $u = 2^x$: $2u^2 - 9u + 4 = 0 \Rightarrow (2u - 1)(u - 4) = 0 \Rightarrow u = 1/2 \Rightarrow x = -1$, or $u = 4 \Rightarrow x = 2$.

3 Simplify the index expression $\frac{(4x^3y^{-2})^{\frac{1}{2}}}{2x^{-1}y^3}$.

a. $x^{\frac{5}{2}}y^{-4}$ ✓

b. $2x^{\frac{5}{2}}y^{-4}$

c. x^2y^{-4}

d. $x^{\frac{5}{2}}y^2$

Numerator: $2x^{3/2}y^{-1}$. Divide by $2x^{-1}y^3$: $x^{3/2-(-1)}y^{-1-3} = x^{5/2}y^{-4}$.

Section 2: Surds and Rationalising Denominators

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4 Rationalise the denominator and simplify $\frac{4+\sqrt{3}}{2-\sqrt{3}}$ into the form $a+b\sqrt{3}$.

a. $11+6\sqrt{3}$ ✓

b. $5+6\sqrt{3}$

c. $11-6\sqrt{3}$

d. $7+4\sqrt{3}$

Multiply top and bottom by $2+\sqrt{3}$: $\frac{(4+\sqrt{3})(2+\sqrt{3})}{4-3}=8+4\sqrt{3}+2\sqrt{3}+3=11+6\sqrt{3}$.

5 Express $(3-2\sqrt{5})^2$ in the simplified form $p+q\sqrt{5}$.

a. $29-12\sqrt{5}$ ✓

b. $29+12\sqrt{5}$

c. $11-12\sqrt{5}$

d. $19-6\sqrt{5}$

$3^2-2(3)(2\sqrt{5})+(2\sqrt{5})^2=9-12\sqrt{5}+20=29-12\sqrt{5}$.

6 Solve the radical equation $\sqrt{2x+7}-x=2$ for all valid real solutions.

a. $x=1$ **only** ✓

b. $x=1$ and $x=-3$

c. $x=3$ only

d. No real solutions

$\sqrt{2x+7}=x+2\Rightarrow 2x+7=x^2+4x+4\Rightarrow x^2+2x-3=0\Rightarrow(x-1)(x+3)=0$. $x=1$ works; $x=-3$ gives $\sqrt{1}-(-3)=4\neq 2$.

Section 3: Remainder and Factor Theorems

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7 When $P(x) = 2x^3 + ax^2 - 5x + 6$ is divided by $(x - 2)$, the remainder is 16. Find the value of a.

a. $a = 1$ ✓

b. $a = -2$

c. $a = 3$

d. $a = 4$

$P(2) = 2(8) + a(4) - 5(2) + 6 = 16 + 4a - 10 + 6 = 4a + 12 = 16 \Rightarrow 4a = 4 \Rightarrow a = 1$.

8 Given that $(2x - 1)$ is a factor of $f(x) = 2x^3 + 3x^2 - 8x + c$, find the constant c.

a. $c = 3$ ✓

b. $c = -3$

c. $c = 6$

d. $c = -2$

$f(1/2) = 2(1/8) + 3(1/4) - 8(1/2) + c = 1/4 + 3/4 - 4 + c = 1 - 4 + c = -3 + c = 0 \Rightarrow c = 3$.

9 Factorise completely the cubic polynomial $g(x) = x^3 - 4x^2 - 7x + 10$.

a. $(x - 1)(x + 2)(x - 5)$ ✓

b. $(x + 1)(x - 2)(x + 5)$

c. $(x - 1)(x - 2)(x - 5)$

d. $(x + 1)(x + 2)(x - 5)$

$g(1) = 1 - 4 - 7 + 10 = 0 \Rightarrow (x - 1)$ is a factor. Dividing gives $x^2 - 3x - 10 = (x + 2)(x - 5)$.

Section 4: The Binomial Theorem

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10 Find the coefficient of x^3 in the binomial expansion of $(1+2x)^6$.

a. 160 ✓

b. 80

c. 240

d. 120

Term is $\binom{6}{3}(1)^3(2x)^3 = 20(8x^3) = 160x^3$. Coefficient is 160.

11 Find the term independent of x (the constant term) in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

a. 240 ✓

b. 160

c. 60

d. 192

General term: $\binom{6}{r}(x^2)^{6-r}(2/x)^r = \binom{6}{r}2^r x^{12-3r}$. Constant when $12-3r=0 \Rightarrow r=4$. Term: $\binom{6}{4}2^4 = 15(16) = 240$.

12 In the expansion of $(1+ax)^5(1-x)$, the coefficient of x is 9. Find the value of a.

a. $a=2$ ✓

b. $a=1$

c. $a=3$

d. $a=-2$

$(1+5ax+\ldots)(1-x) = 1+(5a-1)x+\ldots$. Given $5a-1=9 \Rightarrow 5a=10 \Rightarrow a=2$.

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